GAGANA K

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OBJECTIVE

Computer Science undergraduate with a strong interest in Artificial Intelligence and Machine Learning. Passionate about applying data-driven methods to real-world problems and understanding model behavior and limitations. Seeking opportunities to contribute to AI/ML projects through experimentation, analysis, and research-oriented problem solving.

EDUCATION

B.Tech in Computer Science
RV University, Bangalore
CGPA: 9.55

Pre-University (Science)
RV PU College, Bangalore — 96.5%

School (10th Grade)
Apollo Convent, Bangalore — 99.36%

TECHNICAL SKILLS

Languages: C, Python, Java, SQL
AI / ML: Machine Learning, Deep Learning, Computer Vision, Active Learning, TinyML
Tools: Docker, Kubernetes, Git
Libraries: NumPy, Pandas, Scikit-learn, TensorFlow
Soft Skills: Effective Communication, Teamwork

CERTIFICATIONS

Programming in Modern C++ – NPTEL
Human Computer Interaction – NPTEL
Supervised Machine Learning – DeepLearning.AI (Andrew Ng)
Data Analysis and Visualization with Pandas – Kaggle Learn
Agile Scrum Foundation – Simplilearn SkillUp

LANGUAGES

English, Kannada, Telugu

PROJECTS

IoT-Based Fire and Gas Detection using TinyML

- Built a hardware-focused IoT system for detecting fire and hazardous gas using environmental sensors.
- Implemented an on-device ML/DL-based classification model using TinyML on a microcontroller.
- Performed real-time inference locally without cloud connectivity.

Active Learning for Data-Efficient Model Training

- Implemented active learning strategies to reduce labeled data requirements in machine learning models.
- Studied uncertainty-based sampling and analyzed learning curves under limited data settings.

Deepfake Detection using Feature Engineering and CNNs

- Developed face authenticity detection models using handcrafted features and CNN-based approaches.
- Compared classical machine learning pipelines with deep learning models.

RESEARCH / PAPERS

Early Detection of Alzheimer's Disease using Machine Learning

- Research work focused on ML-based early-stage Alzheimer's detection using healthcare datasets.
- Manuscript submitted to a conference (under review).

EXPERIENCE

Summer Internship – AI in Agriculture

- Analyzed agricultural datasets to study soil carbon content and crop-related patterns.
- Applied data preprocessing and visualization techniques using Python to interpret trends.